Name- Abiy Zelalem Tegegne

ID - 213458

Advances Quantum Theory

Seminar - One (I)

1. Harmonic oscillator

Given

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{x}^2}{2}$$

a) Given

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)$$

$$\hat{a}^{+} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)$$

Requires

Show, $[\hat{a}, \hat{a}^{+}] = 1$

Solution

$$[\hat{a}, \hat{a}^{+}] = \left[\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)\right]$$

$$= \left(\sqrt{\frac{m\omega}{2\hbar}}\right)^2 \left[\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right), \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)\right]$$

$$= \frac{m\omega}{2\hbar}\left(\underbrace{[\hat{x}, \hat{x}]}_{0} - [x, \frac{i\hat{p}}{m\omega}] + [\frac{i\hat{p}}{m\omega}, \hat{x}] + \underbrace{[\frac{i\hat{p}}{m\omega}, \frac{-i\hat{p}}{m\omega}]}_{0}\right)$$

$$= \frac{m\omega}{2\hbar}\left(-\left(\frac{i\hbar}{m\omega}\right)i + \frac{i(-i\hbar)}{m\omega}\right) = \frac{m\omega}{2\hbar}\left(\frac{\hbar}{m\omega} + \frac{\hbar}{m\omega}\right)$$

$$= \frac{2\hbar m\omega}{2\hbar m\omega} = \frac{2\hbar m\omega}{2\hbar m\omega} = 1$$

$$[\hat{a}, \hat{a}^{+}] = 1$$

b) Given

$\hat{N} = \hat{a}^{+}\hat{a}$

Requires

$\hat{H} = \hbar\omega(\hat{N} + \frac{1}{2})$ → show this?

Solution

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^{+}), \quad \hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}^{+} - \hat{a})$$

$$\hat{H} = \frac{1}{2m}\left(i\sqrt{\frac{\hbar m\omega}{2}}\,(\hat{a}^{+} - \hat{a})\right)^2 + \frac{m\omega^2}{2}\left(\sqrt{\frac{\hbar}{2m\omega}}\,(\hat{a} + \hat{a}^{+})\right)^2$$

$$= \frac{1}{2m}\left(-\frac{\hbar m\omega}{2}\left(\hat{a}^{+2} - \hat{a}^{+}\hat{a} - \hat{a}\hat{a}^{+} + \hat{a}^{2}\right)\right) + \frac{m\omega^2}{2}\left(\frac{\hbar}{2m\omega}\left(\hat{a}^{+2} + \hat{a}^{2} + \hat{a}^{+}\hat{a} + \hat{a}\hat{a}^{+}\right)\right)$$

$$= \frac{\hbar\omega}{4}\left(-\hat{a}^{+2} - \hat{a}^{2} + \hat{a}^{+}\hat{a} + \hat{a}\hat{a}^{+}\right) + \frac{\hbar\omega}{4}\left(\hat{a}^{+2} + \hat{a}^{2} + \hat{a}^{+}\hat{a} + \hat{a}\hat{a}^{+}\right)$$

$$= \frac{\hbar\omega}{4}\left(\cancel{-\hat{a}^{+2}} \cancel{-\hat{a}^{2}} + \hat{a}^{+}\hat{a} + \hat{a}\hat{a}^{+} + \cancel{\hat{a}^{+2}} + \cancel{\hat{a}^{2}} + \hat{a}^{+}\hat{a} + \hat{a}\hat{a}^{+}\right)$$

$$= \frac{\hbar\omega}{4}\left(2\hat{a}^{+}\hat{a} + 2\hat{a}\hat{a}^{+}\right) = \frac{\hbar\omega}{2}\left(\hat{a}^{+}\hat{a} + \underbrace{\hat{a}\hat{a}^{+}}_{1 + \hat{a}^{+}\hat{a}}\right)$$

$$= \frac{\hbar\omega}{2}\left(\hat{a}^{+}\hat{a} + 1 + \hat{a}^{+}\hat{a}\right) = \frac{\hbar\omega}{2}\left(2\hat{a}^{+}\hat{a} + 1\right) = \hbar\omega\left(\underbrace{\hat{a}^{+}\hat{a}}_{\hat{N}} + \frac{1}{2}\right)$$

$$\hat{H} = \hbar\omega\left(\hat{N} + \frac{1}{2}\right)$$

c) Given

$\hat{N}|n\rangle = n|n\rangle$

Required

$[\hat{N}, \hat{a}^{+}] = ?$
$[\hat{N}, \hat{a}] = ?$
$\hat{N}\hat{a}^{+}|n\rangle = ?$
$\hat{N}\hat{a}|n\rangle = ?$

Solution

$$[\hat{N}, \hat{a}^{+}] = [\hat{a}^{+}\hat{a}, \hat{a}^{+}] = \hat{a}^{+}\underbrace{[\hat{a}, \hat{a}^{+}]}_{=1} + \underbrace{[\hat{a}^{+}, \hat{a}^{+}]}_{=0}\hat{a}$$
$$= \hat{a}^{+}$$

$$[\hat{N}, \hat{a}] = [\hat{a}^{+}\hat{a}, \hat{a}] = \hat{a}^{+}\underbrace{[\hat{a}, \hat{a}]}_{=0} + \underbrace{[\hat{a}^{+}, \hat{a}]}_{=-1}\hat{a}$$
$$= -\hat{a}$$

1st – C1 – C.

(C2) - Give an interpretation what happens to the eigenstates of the number operator $|n\rangle$, when applying the annihilation [Page-2] or creation operator,

Solution

$$\left.\begin{aligned}\hat{a}|n\rangle &= \sqrt{n}\,|n-1\rangle \\ \hat{a}^{+}|n\rangle &= \sqrt{n+1}\,|n+1\rangle\end{aligned}\right\}$$

For creation operator ($\hat{a}^{+}$)

✓ Eigen values increase by 1

✓ For eigen state add/create one quantum particle/energy

$$\underbrace{n \longrightarrow n+1}$$

For Annihilation operator

Eigenstate remove/reduce one quantum particle/energy

$$\underbrace{n \longrightarrow n-1}$$

Q3 - Given

$$\left.\begin{aligned}\hat{N}|n\rangle &= n|n\rangle \\ [\hat{N}, \hat{a}^{+}] &= \hat{a}^{+} \\ [\hat{N}, \hat{a}] &= -\hat{a}\end{aligned}\right\}$$

Requires

$\hat{N}\hat{a}^{+}|n\rangle = ?$

$\hat{N}\hat{a}|n\rangle = ?$

Solution

$$\begin{aligned}\hat{N}\hat{a}^{+}|n\rangle &= (\hat{a}+\hat{a}^{+}N)|n\rangle \quad \text{, where, } \hat{N}\hat{a}^{+} = \hat{a}^{+}\hat{N} + \hat{a}^{+} \\ &= (\hat{a}^{+}+\hat{a}^{+}\hat{N})|n\rangle \\ &= \hat{a}^{+}|n\rangle + \hat{a}^{+}\hat{N}|n\rangle \\ &= \hat{a}^{+}|n\rangle + \hat{a}^{+}n|n\rangle \\ &= \underline{\underline{(n+1)\,\hat{a}^{+}|n\rangle}} = \underline{\underline{(n+1)(\sqrt{n+1})\,|n+1\rangle}}\end{aligned}$$

$$\begin{aligned}\hat{N}\hat{a}|n\rangle &= (\hat{a}\hat{N} - \hat{a})|n\rangle \quad \text{, where, } \hat{N}\hat{a} = \hat{a}\hat{N} - \hat{a} \\ &= \hat{a}\hat{N}|n\rangle - \hat{a}|n\rangle \\ &= \hat{a}\,n|n\rangle - \hat{a}|n\rangle \\ &= \underline{\underline{(n-1)\,\hat{a}|n\rangle}} = \underline{\underline{(n-1)\sqrt{n}\,|n-1\rangle}}\end{aligned}$$

C4 - Based on the above solution what I have don

Given

$\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle$
$\hat{a}^{+}|n\rangle = \sqrt{n+1}\,|n+1\rangle$

$\hat{N}\hat{a}^{+}|n\rangle = (n+1)(\sqrt{n+1})\,|n+1\rangle$
$\hat{N}\hat{a}|n\rangle = (n-1)\sqrt{n}\,|n-1\rangle$

Required

is still an eigenstate of the number operator?

Solution

Appling the creation or aninilation operator keeps the state

1- the eigen state is same
2- Eigen value is changes

1 $\hat{a}^{+}$ (Creation) $\qquad$ $\hat{N}\hat{a}^{+}$ (Creation)

$\hat{a}^{+} \to n \to n+1$ - eigen value $\qquad$ $n \to (n+1)\sqrt{n+1} \to$ eigen value

$|n\rangle - |n+1\rangle$ - eigen state $\qquad$ $|n\rangle \to |n+1\rangle \to$ eigen state

Same eigen state but different eigen value when we apply ($\hat{a}^{+}$ and $\hat{N}\hat{a}^{+}$) the same for ($\hat{a}$ & $\hat{N}\hat{a}$)

2. Qubit Basics

Given

$|\psi\rangle = \frac{1}{N}\left(\sqrt{3}|0\rangle + i|1\rangle\right)$

Required

a) Normalize $|\psi\rangle$ i.e calculate N?

b) $X = |+\rangle\langle +| - |-\rangle\langle -|$, found $P(+) = ?$

c) found angle θ and ϕ at ψ of Bloch sphere?

d) $|\psi\rangle \otimes |+\rangle = ?$

Solution

a) $\langle \psi|\psi\rangle = 1$, Where, $|\psi\rangle = \frac{1}{N}\left(\sqrt{3}|0\rangle + i|1\rangle\right)$

$\langle\psi| = (|\psi\rangle)^{*} = \frac{1}{N}\left(\langle 0|\sqrt{3} - i\langle 1|\right)$

$$\langle\psi|\psi\rangle = \frac{1}{N}\left(\sqrt{3}\langle 0| - i\langle 1|\right)\left(\frac{1}{N}\left(\sqrt{3}|0\rangle + i|1\rangle\right)\right)$$

$$= \frac{1}{N^2}\left((\sqrt{3})^2 \underbrace{\langle 0|0\rangle}_{=1} + \sqrt{3}\, i \underbrace{\langle 0|1\rangle}_{=0} - \sqrt{3}\, i \underbrace{\langle 1|0\rangle}_{=0} - i^2 \underbrace{\langle 1|1\rangle}_{=1}\right)$$

$$= \frac{1}{N^2}\left(3 - (-1)\right) = \frac{4}{N^2}$$

$\langle\psi|\psi\rangle = \frac{4}{N^2}$, where $\langle\psi|\psi\rangle = 1$

$1 = \frac{4}{N^2} \Rightarrow N^2 = 4 \Rightarrow N = \sqrt{4}$, $N = 2$

$\underline{N = 2}$

b) <u>Given</u>

$K = |+\rangle\langle +| - |-\rangle\langle -|$

$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$

$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

<u>Requires</u>

$P(+) = ?$

<u>Solution</u>

Apply Born rule, $P(+) = |\langle +|\psi\rangle|^2$

$$P(+) = |\langle +|\psi\rangle|^2 = \left|\left(\frac{1}{\sqrt{2}}(\langle 0| + \langle 1|)\right)\left(\frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle\right)\right|^2$$

$$= \left|\frac{\sqrt{3}}{2\sqrt{2}}\underbrace{\langle 0|0\rangle}_{=1} + \frac{i}{2\sqrt{2}}\underbrace{\langle 0|1\rangle}_{0} + \frac{\sqrt{3}}{2\sqrt{2}}\underbrace{\langle 1|0\rangle}_{=0} + \frac{i}{2\sqrt{2}}\underbrace{\langle 1|1\rangle}_{=1}\right|^2$$

$$= \left|\frac{\sqrt{3}}{2\sqrt{2}} + 0 + 0 + \frac{i}{2\sqrt{2}}\right|^2 = \left|\frac{\sqrt{3}}{2\sqrt{2}} + \frac{i}{2\sqrt{2}}\right|^2$$

$$= \left(\sqrt{\left(\frac{\sqrt{3}}{2\sqrt{2}}\right)^2 + \left(\frac{1}{2\sqrt{2}}\right)^2}\right)^2$$

$$= \left(\sqrt{\frac{3}{8} + \frac{1}{8}}\right)^2 = \left(\sqrt{\frac{4}{8}}\right)^2 = \left(\sqrt{\frac{1}{2}}\right)^2 = \frac{1}{2}$$

$\underline{\underline{P(+) = \frac{1}{2}}}$, where, $\left.\begin{array}{l} P(+) = \frac{1}{2} \\ P(-) = \frac{1}{2} \end{array}\right\}$ for the Pure state

c) Given $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$

Required

$\theta = ?$

$\phi = ?$

$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$

Solution

Let's start from real part of quantum state

$\frac{\sqrt{3}}{2}|0\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle$

Imaginary part

$\frac{i}{2}|1\rangle = e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$

$= \left\{ \cos\left(\frac{\theta}{2}\right) = \frac{\sqrt{3}}{2}, \quad e^{i\phi}\sin\left(\frac{\theta}{2}\right) = \frac{i}{2} \right\}$

There for let's go to the full mathematical calculation

Let's first calculate "θ"

$\cos\left(\frac{\theta}{2}\right) = \frac{\sqrt{3}}{2}$

$\frac{\theta}{2} = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\theta = 2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

where, $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$

then,

$\theta = 2 \cdot \frac{\pi}{6}$

$\theta = \frac{\pi}{3}$

Let's calculate ϕ

$e^{i\phi}\sin\left(\frac{\theta}{2}\right) = \frac{i}{2}$

$e^{i\phi} \cdot \sin\left(\frac{\pi/3}{2}\right) = \frac{i}{2}$

$e^{i\phi}\sin\left(\frac{\pi}{6}\right) = \frac{i}{2}$

$e^{i\phi} \cdot \frac{1}{2} = \frac{i}{2}$, where $\boxed{\sin(30) = \frac{1}{2}}$

$e^{i\phi} = i$

where, $\boxed{e^{i\phi} = \cos\phi + i\sin\phi}$

$e^{i\phi} = i \Rightarrow \cos\phi + i\sin\phi = i$

$\left.\begin{array}{l}\cos\phi = 0 \\ i\sin\phi = i\end{array}\right\}$ imaginary and Real part separates

$\cos\phi = 0$

$\phi = \cos^{-1}(0) \Rightarrow \phi = \frac{\pi}{2}$

Therefore
$$\theta = \frac{\pi}{3}, \quad \phi = \frac{\pi}{2}$$

2) Given

$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{i}{2}|1\rangle$

$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Required

$|\psi\rangle \otimes |+\rangle = ?$

Solution

$$|\psi\rangle \otimes |+\rangle = \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{i|1\rangle}{2}\right) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
$$= \frac{1}{2\sqrt{2}}\left(\sqrt{3}|00\rangle + \sqrt{3}|01\rangle + i|10\rangle + i|11\rangle\right)$$

Standard basis: $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

3 Time-independent perturbation

Given

$H_0 = \frac{p^2}{2m} - \frac{e^2}{r}$ - non-relativistic Hamiltonian

$H' = \sqrt{m^2c^4 + p^2c^2} - \frac{e^2}{r}$ -

a) Required

Calculated the H_0 deviate (perturbed term) = ?

Solution

$$H' = \sqrt{m^2c^4 + p^2c^2} - \frac{e^2}{r} = \sqrt{m^2c^4\left(1 + \frac{p^2}{m^2c^2}\right)} - \frac{e^2}{r}$$
$$= mc^2\sqrt{\left(1 + \frac{p^2}{m^2c^2}\right)} - \frac{e^2}{r}$$

Where

$$x = \frac{p^2}{m^2c^2}$$

$$H' = mc^2\sqrt{1+\frac{p^2}{m^2c^2}} - \frac{e^2}{r}$$ } To Solve $\sqrt{1+\frac{p^2}{m^2c^2}}$ we uses Taylor series

Taylor series

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + O(x^3$$

$$H' = mc^2\sqrt{1+\frac{p^2}{m^2c^2}} - \frac{e^2}{r} = mc^2\left(1 + \frac{p^2}{2m^2c^2} - \frac{p^4}{8m^3c^2} - \frac{e^2}{r}\right)$$

$$H' = mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} - \frac{e^2}{r} = \underbrace{mc^2}_{\text{constant}} + \underbrace{\left(\frac{p^2}{2m} - \frac{e^2}{r}\right)}_{H_0} + \underbrace{\left(\frac{-p^4}{8m^3c^2}\right)}_{V}$$

$$H' = mc^2 + H_0 + V$$

$$H' - H_0 = mc^2 + V \Rightarrow \text{(Remove the constant } mc^2\text{)}$$

$$V = H' - H_0 = -\frac{p^4}{8m^3c^2}$$

$$\boxed{V = -\frac{p^4}{8m^3c^2} \rightarrow}$$ $\rightarrow$ The Perturbed terms.

b) Given

$$\langle \tfrac{1}{r} \rangle_{100} = \frac{1}{a_0}$$

$$\langle \tfrac{1}{r^2} \rangle_{100} = \frac{2}{a_0^2}$$

$$E_{100} = \frac{-e^2}{2a_0}$$

Requires

$$\Delta E^{(1)}_{100} = ?$$

Solution

$$\Delta E^{(1)}_{nlm} = \langle nlm | V | nlm \rangle$$ - - - - - - - - → General equation

$nlm = 100$ - - - - (x)

$V = -\dfrac{p^4}{8m^3c^2}$ - - - (xx)

(x) $\Delta E^{(1)}_{100} = \langle 100 | -\dfrac{p^4}{8m^3c^2} | 100 \rangle = -\dfrac{1}{8m^3c^2} \langle 100 | p^4 | 100 \rangle$

we used p^4 in the term of eigenvalue equation

$$H_0 = \frac{p^2}{2m} - \frac{e^2}{r}$$

- - - - - - - - - - - - xxx

$H_0 |\psi\rangle = E |\psi\rangle$ - - - -

where,

$$\left(\frac{p^2}{2m} - \frac{e^2}{r}\right)|\psi\rangle = E|\psi\rangle$$

$$\frac{p^2}{2m}|\psi\rangle = E|\psi\rangle + \frac{e^2}{r}|\psi\rangle \Rightarrow p^2|\psi\rangle = 2m\left(E + \frac{e^2}{r}\right)|\psi\rangle$$

$$p^2|\psi\rangle = 2m\left(E + \frac{e^2}{r}\right)|\psi\rangle$$

Apply p^4 on the state $|\psi\rangle$

$$p^4|\psi\rangle = 4m^2\left(E + \frac{e^2}{r}\right)^2|\psi\rangle$$

$$= 4m^2\left(E^2 + \frac{2Ee^2}{r} + \frac{e^4}{r^2}\right)|\psi\rangle$$

$$p^4|\psi\rangle = 4m^2\left(E^2 + \frac{2Ee^2}{r} + \frac{e^4}{r}\right)|\psi\rangle$$

To solve this we must take the expectation of $\langle p^4 \rangle$ | page-9 |

$$\langle p^4 \rangle = 4m^2\left(E^2 + 2Ee^2\langle \frac{1}{r} \rangle + e^4 \langle \frac{1}{r^2} \rangle \right)$$

The ground state hydrogen element for $|100\rangle$ 1st state

$$E_{100} = \frac{-e^2}{2a_0}, \quad \langle \frac{1}{r} \rangle_{100} = \frac{1}{a_0}, \quad \langle \frac{1}{r^2} \rangle_{100} = \frac{2}{a_0^2}$$

Evaluate $\langle p^4 \rangle_{100}$

$$\langle p^4 \rangle_{100} = 4m^2\left(E_{100}^2 + 2E_{100}e^2\langle \frac{1}{r} \rangle_{100} + e^4 \langle \frac{1}{r^2} \rangle_{100} \right)$$

$$= 4m^2\left(\frac{e^4}{4a_0^2} - \cancel{E_{100}} \frac{e^4}{a_0^2} + \frac{2e^4}{a_0^2} \right)$$

$$= \cancel{4}m^2 \left(\frac{e^4 - 4e^4 + 8e^4}{\cancel{4}a_0^2} \right) = \frac{5e^4 m^2}{a_0^2}$$

$$\langle p^4 \rangle_{100} = \frac{5e^4}{a_0^2} m^2$$

$$\Delta E_{100}^{(r)} = -\frac{1}{8m^3c^2} \langle 100 | p^4 | 100 \rangle = -\frac{1}{8m^3c^2}\left(\frac{5e^4}{a_0^2} m^2 \right) = -\frac{5e^4}{8mc^2a_0^2}$$

$$\Delta E_{100}^{(r)} = \frac{-5e^4}{8mc^2a_0^2}$$

If necessary $E_{100} = \frac{-e^2}{2a_0}$

$$\Delta E_{100}^{(r)} = -\frac{5}{2mc^2} E_{100}^2$$

c) Extra, given

$$\psi_{100}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-\frac{|\vec{r}|}{a_0}},$$

Required

$\langle \frac{1}{r} \rangle = ?$

Solution $\langle \frac{1}{r} \rangle = \int \psi^*_{100}(\vec{r}) \frac{1}{r} \psi_{100}(\vec{r}) \, d^3r$

where, $d^3r = r^2 \sin\theta \, dr \, d\theta \, d\phi$

$\psi^*_{100}(\vec{r}) \psi_{100}(\vec{r}) = |\psi|^2 = \dfrac{e^{-\frac{2r}{a_0}}}{\pi a_0^3} \Rightarrow |\psi|^2 = \dfrac{e^{-\frac{2r}{a_0}}}{\pi a_0^3}$

$$\langle \frac{1}{r} \rangle = \int_0^\infty \int_0^\pi \int_0^{2\pi} \frac{e^{-2r/a_0}}{\pi a_0^3 \cdot r} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \frac{1}{\pi a_0^3} \int_0^{2\pi} d\phi \int_0^\pi \sin\theta \, d\theta \int_0^\infty \frac{r e^{-2r/a_0}}{\cancel{\phi}} dr$$

where, the Angular terms

$$\int_0^{2\pi} d\phi \int_0^\pi \sin\theta \, d\theta = \phi \Big|_0^{2\pi} \left(-\cos\theta\right) \Big|_0^\pi$$

$$= (2\pi - 0)(-)\left(\cos(\pi) - \cos(0)\right)$$

$$= 2\pi(-)(-1-1) = 2\pi(-)(-2)$$

$$= 2\pi \times 2$$

$$= 4\pi$$

The radial terms

$$\int_0^\infty \frac{r e^{-2r/a_0}}{\cancel{\phi}} dr = \int_0^\infty r e^{-\frac{2r}{a_0}} dr$$

Apply Integration by parts

$u = r$, $\dfrac{du}{dr} = \dfrac{d(r)}{dr} \Rightarrow \dfrac{du}{dr} = 1 \Rightarrow du = dr$

$$dv = e^{-\frac{2r}{a_0}}dr$$

$$\int_0^v dv = \int_0^\infty e^{-\frac{2r}{a_0}}dr \Rightarrow v = \frac{e^{-\frac{2r}{a_0}}}{-\frac{2}{a_0}}$$

$$\boxed{v = -\frac{a_0}{2}e^{-2r/a_0}} \text{ - - - - - - - - - - } \alpha x$$

Apply the formula

$$\int u\,dv = uv - \int v\,du$$

So,

$$\int_0^\infty re^{-2r/a_0}dr = \left[-\frac{a_0}{2}re^{-\frac{2r}{a_0}}\right]_0^\infty - \int_0^\infty -\frac{a_0}{2}e^{-\frac{2r}{a_0}}dr$$

$$= \left[-\frac{a_0}{2}re^{-2r/a_0}\right]_0^\infty + \frac{a_0}{2}\int_0^\infty e^{-2r/a_0}dr$$

$$\int_0^\infty re^{-2r/a_0}dr = \underbrace{\left[-\frac{a_0 \infty}{2}\frac{1}{e^{-\frac{2\infty}{a_0}}} - \frac{a_0 \cdot 0\, e^{-2r/a_0}}{2}\right]}_{0} + \frac{a_0}{2}\int_0^\infty e^{-2r/a_0}dr$$

$$\int_0^\infty re^{-2r/a_0}dr = \frac{a_0}{2}\int_0^\infty e^{-2r/a_0}dr$$

$$= \frac{\frac{a_0}{2}}{-\frac{2}{a_0}}\left[e^{-\frac{2r}{a_0}}\right]_0^\infty = -\frac{a_0^2}{4}\left(e^{-\frac{2\infty}{a}} - e^{-\frac{2\cdot 0}{a_0}}\right)$$

$$= -\frac{a_0^2}{4}\left[\underbrace{e^{-\frac{2\infty}{a_0}}}_{0} - \underbrace{e^{-\frac{2\cdot 0}{a_0}}}_{1}\right]$$

$$= -\frac{a_0^2}{4}$$ //

$$\int_0^\infty r e^{-\frac{2r}{a_0}} dr = \frac{a_0^2}{4}$$

$$\int_0^{2\pi} d\phi \int_0^\pi \sin\theta \, d\theta = 4\pi$$

$$\langle \tfrac{1}{r} \rangle = \frac{1}{\pi a_0^3} \underbrace{\int_0^{2\pi} d\phi \int_0^\pi \sin\theta \, d\theta}_{4\pi} \underbrace{\int_0^\infty r e^{-\frac{2r}{a_0}} dr}_{\frac{a_0^2}{4}}$$

$$\langle \tfrac{1}{r} \rangle = \frac{4\pi}{\pi a_0^3} \cdot \frac{a_0^2}{4} = \frac{1}{a_0}$$

$$\langle \tfrac{1}{r} \rangle = \frac{1}{a_0} \Rightarrow$$

$$\langle \tfrac{1}{r} \rangle = a_0$$